

Lesson 5-4 · Period 1

Solving Radical Equations — Foundation

Klimesara-adapted · Student-centered · No Blooket

DOK 2

5 min

notice & wonder

Explore & Reason · Launch problem **A.** Solve $3(a + 1)^2 + 2 = 11$. Use at least two methods.

B. Try each method on $3\sqrt{a + 1} + 2 = 11$.

C. Which method works better with a radical? What problems arise with the other?

DOK 1-2

12 min

teacher models

Isolate–Raise–Check Rules (keep on your page)

1. **Isolate** the radical on one side of the equation.
2. **Raise** both sides to the index power ($\sqrt{}$ $\rightarrow$ square; $\sqrt[3]{}$ $\rightarrow$ cube).
3. Solve the resulting polynomial equation.
4. **Check** each candidate in the *original* equation — reject any extraneous solutions.

COMMON ERROR: Squaring is NOT reversible — always substitute into the ORIGINAL.

Explore — Think-Pair-Share (33 min)

Algebra 2 · Lesson 5-4 · Period 1

DOK 1-2

33 min

student work

6 items in order. Plan ~5–6 min per item. Cut #4 if pace slows.

Try It 1 · Isolate and raise to solve $\sqrt{x-2} + 3 = 5$ and $\sqrt[3]{x-1} = 2$

Try It 3 · Solve + check for extraneous $x = \sqrt{7x+8}$ and $x+2 = \sqrt{x+2}$

Practice #22 + #8 · Solve radical equations $\sqrt{4x} = 11$ then $\sqrt{8x+9} =$
 x — substitute into ORIGINAL to check

Practice #15 · Identify extraneous (generalize) Explain how to identify an

DOK 2

5 min

academic conversation

Sentence frame · Pair share “I isolated the radical to get _____. Squaring both sides gives _____. My candidate solutions are _____. When I check in the ORIGINAL equation, _____ is valid and _____ is extraneous because _____.”

Bridge to Period 2 Next class: equations with two radicals or a radical and a binomial.

Same Isolate–Raise–Check logic — more complex setup.

DOK 1-2

2 min

not graded

Complete on your exit ticket

1. To solve a radical equation I first _____.
2. I raise both sides to power _____ because _____.
3. An extraneous solution happens when _____.